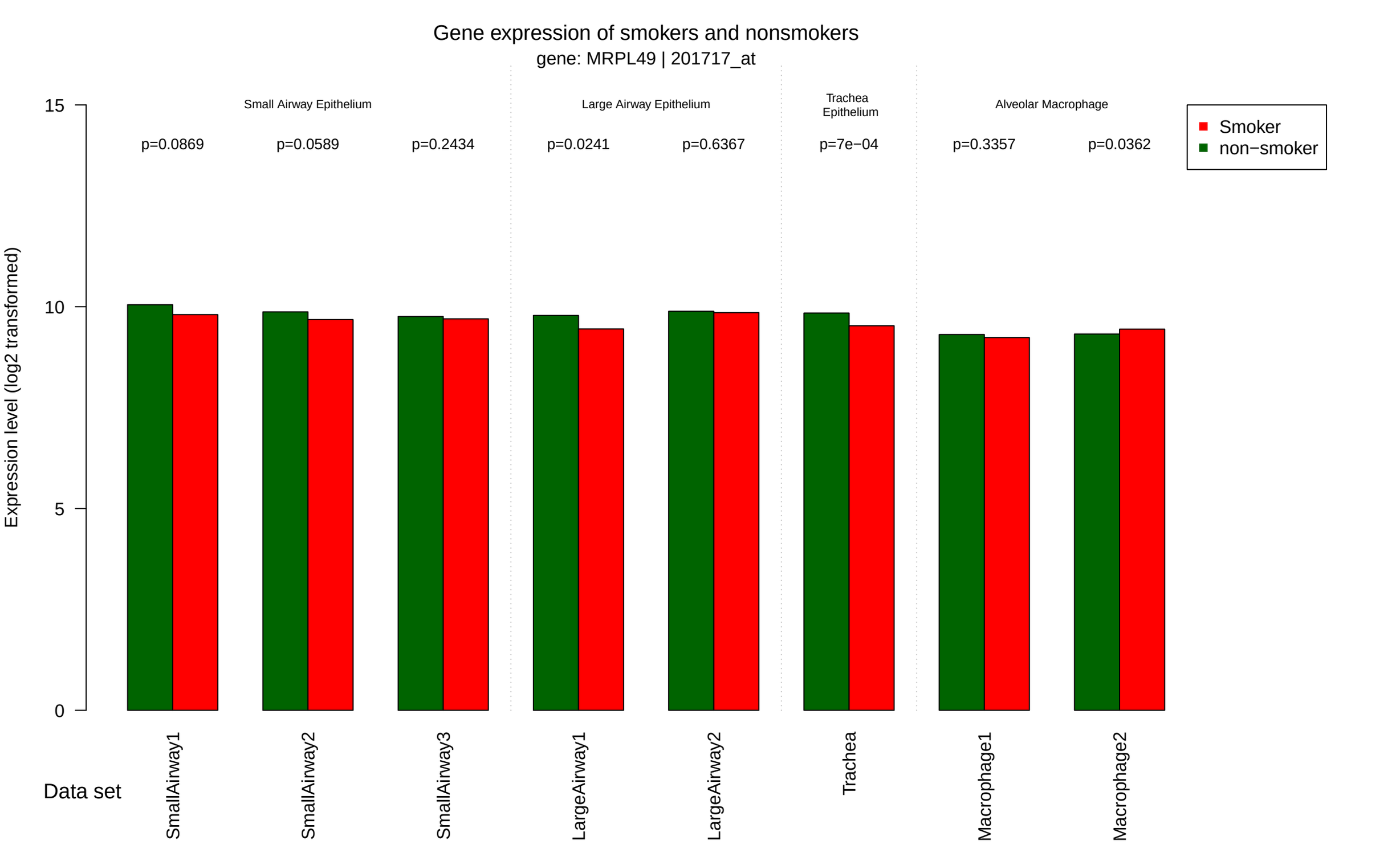
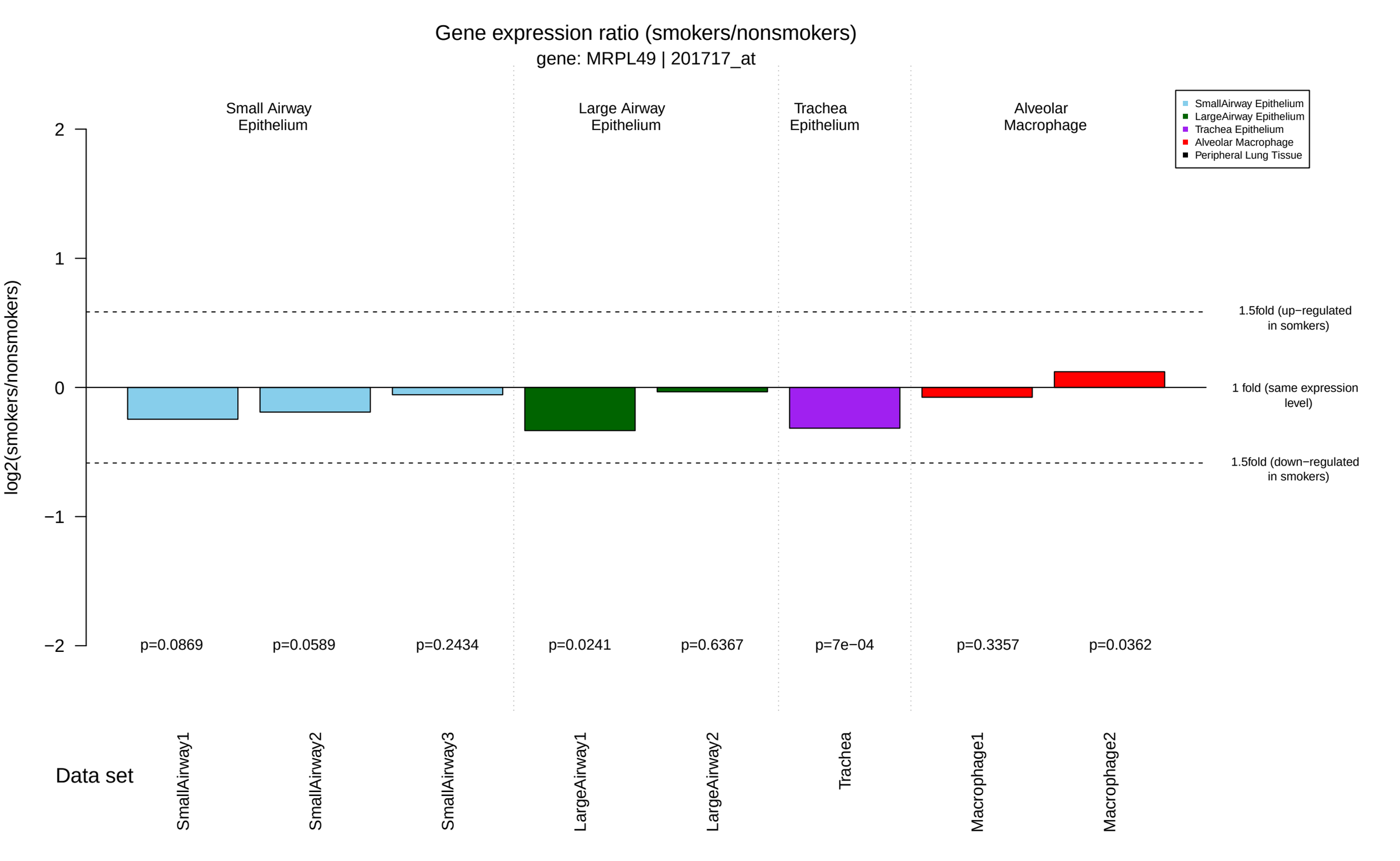
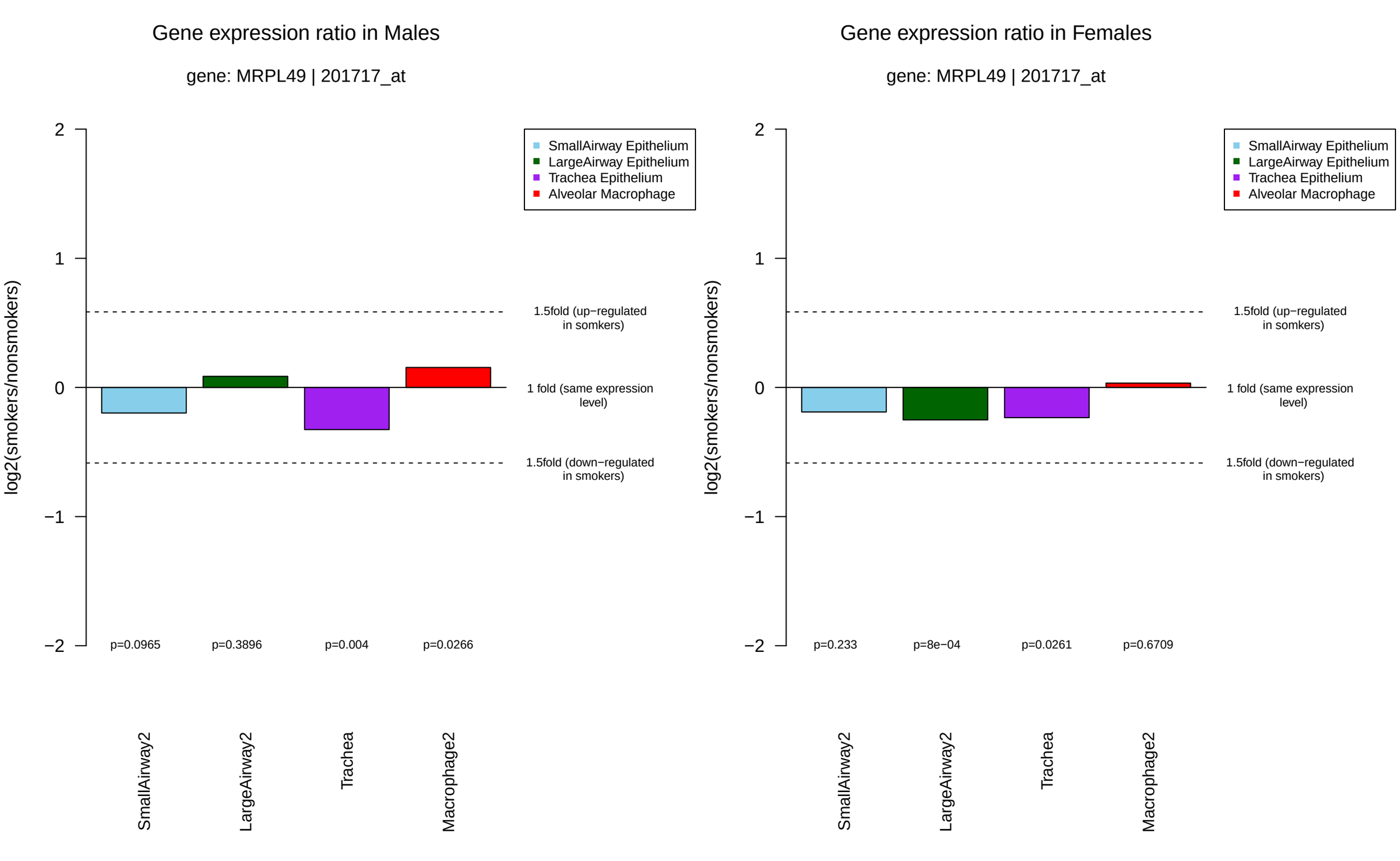
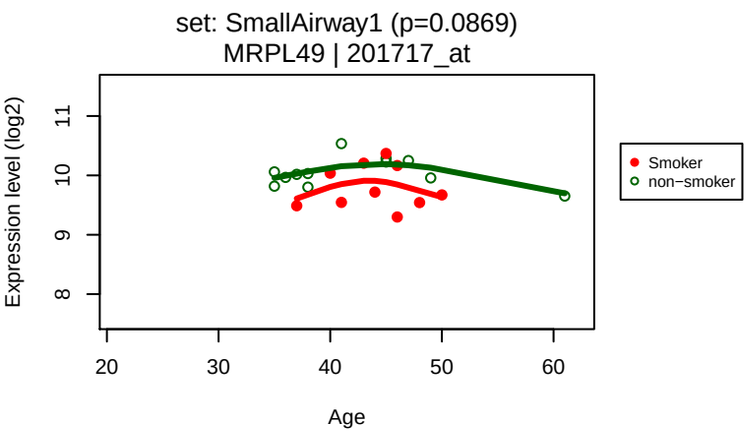
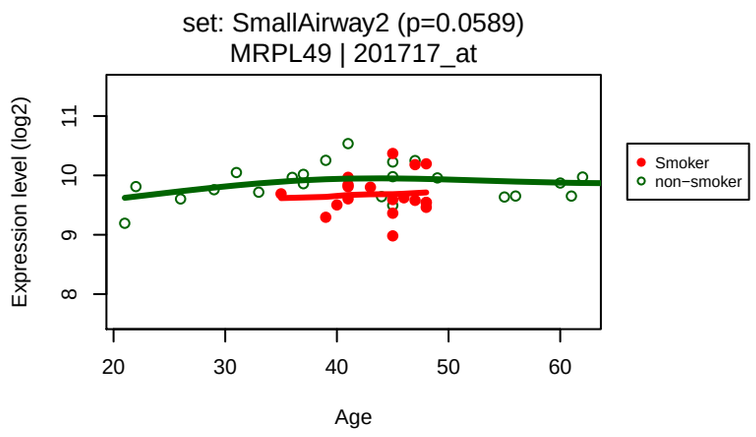
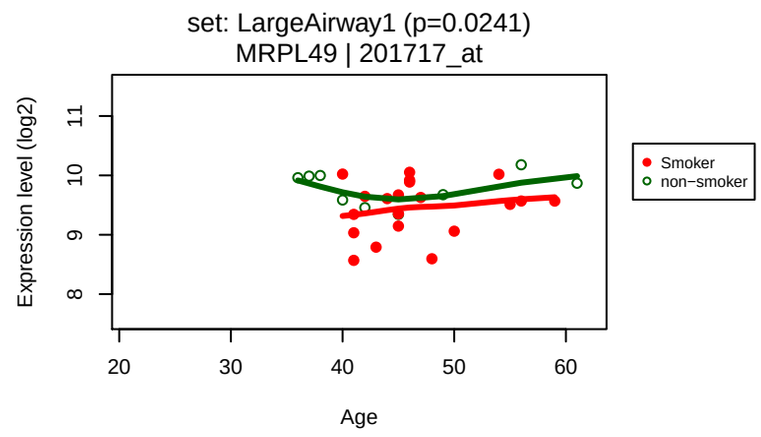

Gene expression of smokers and nonsmokers
gene: MRPL49 | 201717_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0869
p=0.0589
p=0.2434
p=0.0241
p=0.6367
p=7e−04
p=0.3357
p=0.0362
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

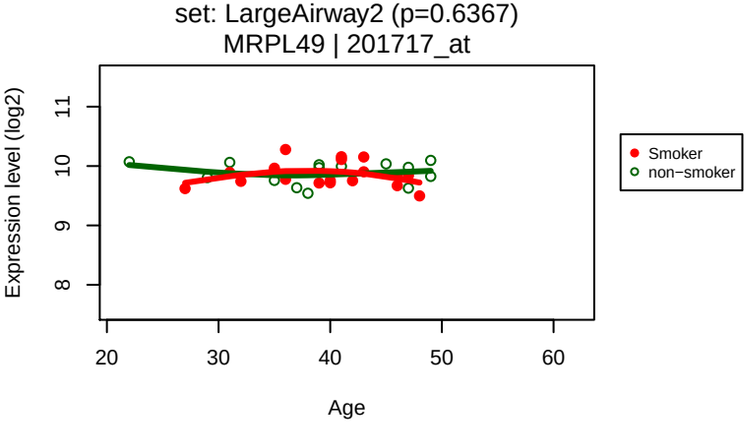

Gene expression ratio (smokers/nonsmokers)
gene: MRPL49 | 201717_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0869
p=0.0589
p=0.2434
p=0.0241
p=0.6367
p=7e−04
p=0.3357
p=0.0362
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

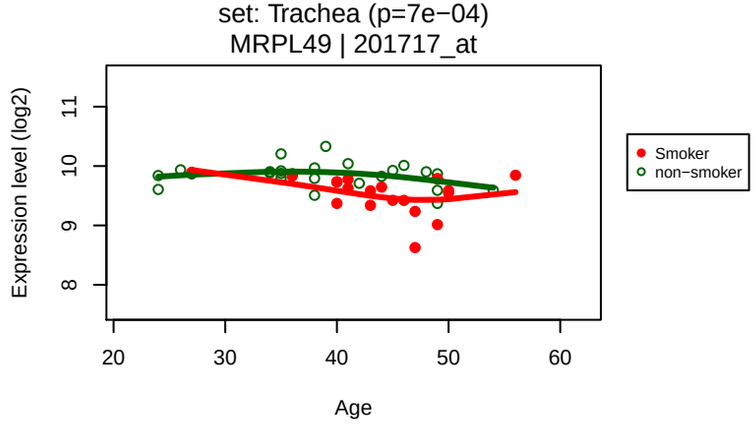

Gene expression ratio in Males
gene: MRPL49 | 201717_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0965
p=0.3896
p=0.004
p=0.0266
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL49 | 201717_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.233
p=8e−04
p=0.0261
p=0.6709
SmallAirway2
LargeAirway2
Trachea
Macrophage2

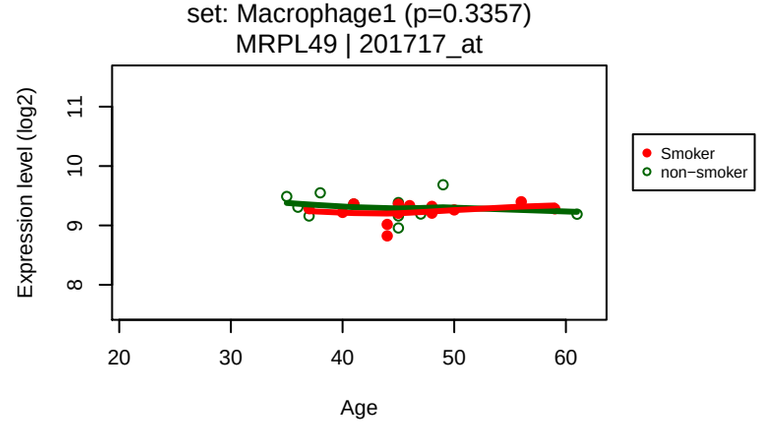

set: SmallAirway1 (p=0.0869)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker

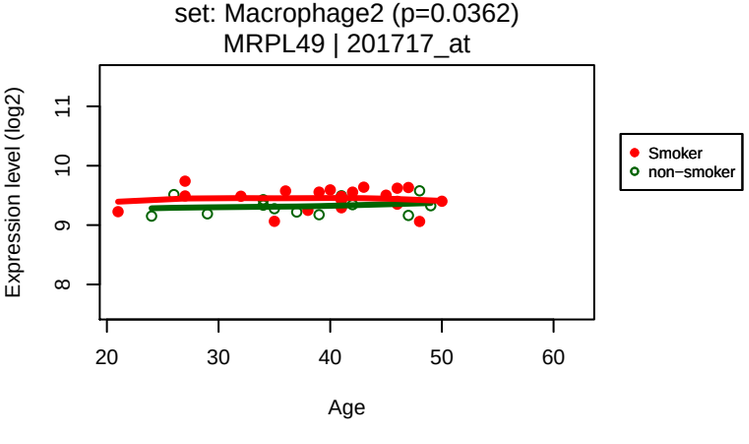

set: SmallAirway2 (p=0.0589)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker

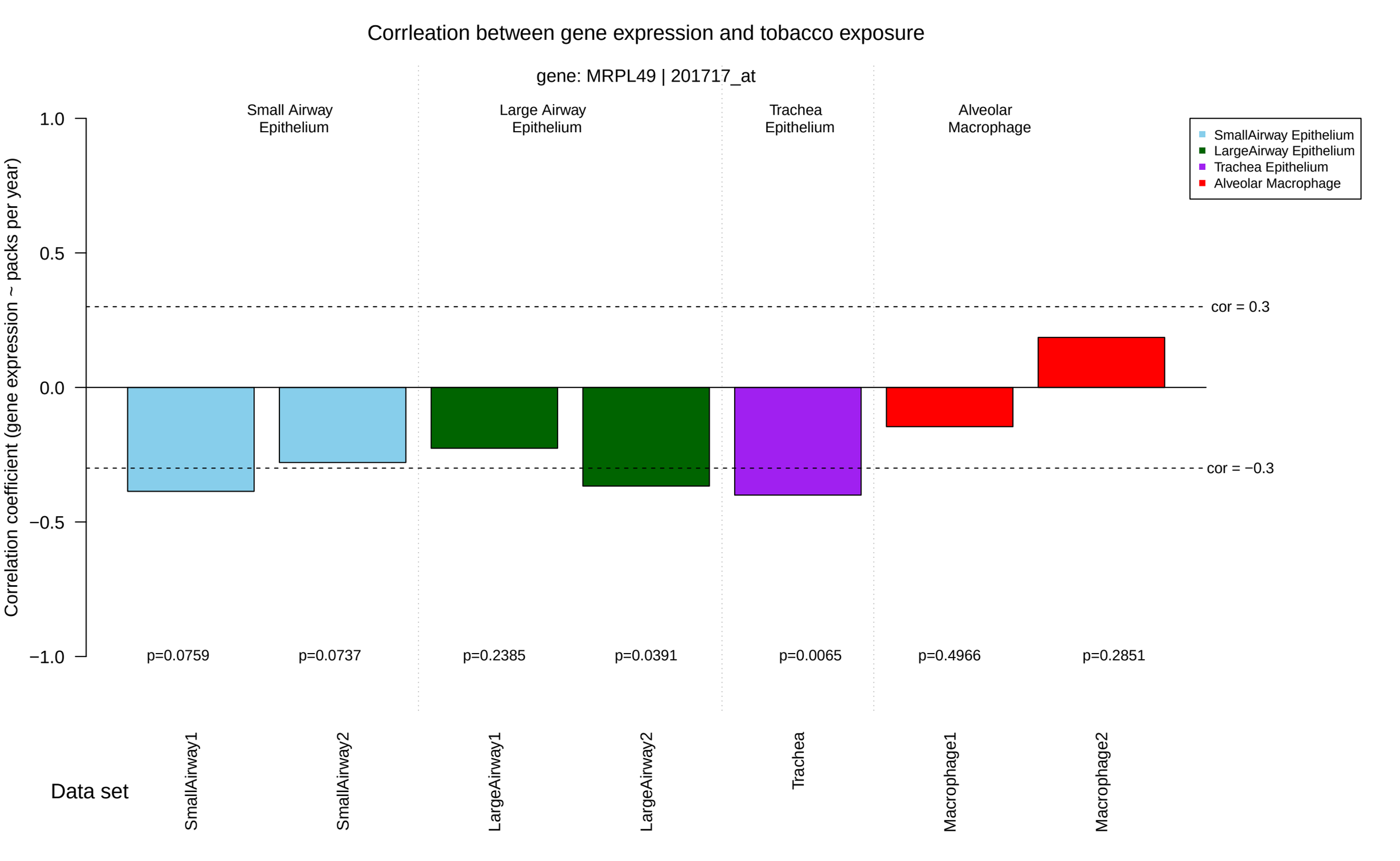

set: LargeAirway1 (p=0.0241)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6367)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3357)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0362)
MRPL49 | 201717_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL49 | 201717_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0759
p=0.0737
p=0.2385
p=0.0391
p=0.0065
p=0.4966
p=0.2851
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2